POWER PLANT ENGINEERING

Subject Code: 23A03702e

UNIT II: Steam Power Plant: Boilers, Fuel Handling and Combustion

1. Steam Power Plant – Introduction

- A ***steam power plant (thermal station)*** converts the chemical energy of fuel (coal) into heat -> steam -> mechanical work -> electricity, using the ***Rankine cycle***.
- Four main circuits: (1) ***coal and ash circuit***, (2) ***air and flue gas circuit***, (3) ***feed water and steam circuit***, (4) ***cooling water circuit***.

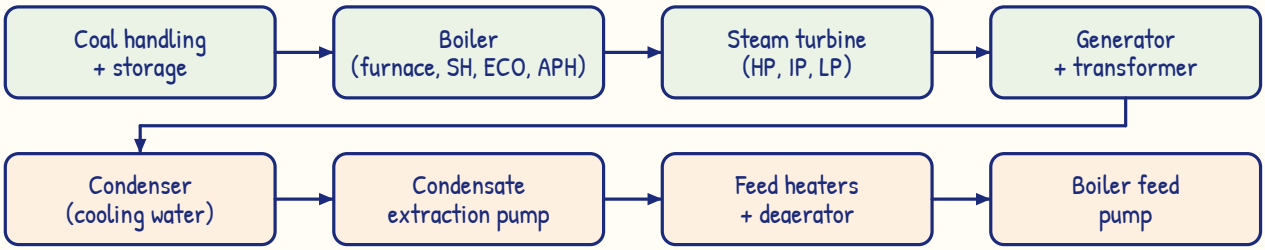


Fig: Layout of a steam power plant

- Flue gas path: furnace -> superheater -> reheater -> ***economiser*** -> ***air pre-heater*** -> dust collector (ESP) -> ***ID fan*** -> chimney.
- Advantages: large power, fuel is cheap, reliable, less initial cost than hydro. Disadvantages: pollution, high running cost, long start-up, needs lot of water.

2. Boilers – Introduction

- ***Boiler (steam generator)*** = closed vessel in which water is converted to steam at constant pressure by heat of combustion.

Classification

Basis	Types
Tube contents	***Fire-tube*** (hot gas inside tubes: Lancashire, Cochran, locomotive) and ***Water-tube*** (water inside tubes: Babcock-Wilcox, Stirling)
Circulation	Natural, forced (La Mont), once-through (Benson, Sulzer)
Pressure	Low (< 20 bar), medium (20–80), high (> 80 bar), supercritical (> 221 bar)

Basis	Types
Use	Stationary (power), portable, marine, locomotive

- Power stations use **water-tube boilers** -- large capacity, high pressure, quick steaming, but need good feed-water quality.

Fire-tube vs water-tube

- Fire-tube: low pressure (<= 25 bar), low output, simple, cheap, large water content, less sensitive to water quality.
- Water-tube: high pressure (up to 300+ bar), large output, high efficiency, small water content, lighter, quick start, needs pure feed water.

Boiler mountings and accessories

- **Mountings** (safety): safety valve, water-level gauge, pressure gauge, stop valve, feed check valve, blow-off cock, fusible plug, manhole.
- **Accessories** (efficiency): **economiser** (heats feed water with flue gas), **air pre-heater**, **superheater**, **reheater**, feed pump, injector, soot blower.

3. Modern High-Pressure and Supercritical Boilers

- Raising pressure and temperature of steam raises Rankine efficiency. Modern drums use natural / controlled circulation up to ~ 180 bar.
- **High-pressure boilers** (above 80 bar) use forced circulation and small-bore tubes; examples: **La Mont**, **Velox**, **Benson**, **Loeffler**.

La Mont boiler (forced circulation)

- A **centrifugal pump** circulates water from the drum through the evaporator tubes at about **8-10 times** the steam generated -> good heat transfer, no steam blanketing. Working pressure ~ 120-150 bar.

Benson boiler (once-through)

- No drum. Feed water passes in one pass through economiser -> evaporator -> superheater; works at **critical / supercritical pressure** (> 225 bar), so water changes to steam without bubbling (no latent heat phase).
- Efficient, light weight, quick start; needs very pure water, no blow-down, complex control.

Loeffler and Velox

- **Loeffler:** steam, not water, circulated (by pump) through the evaporator drum; avoids scale. **Velox:** supercharged combustion (gas turbine compressor drives gas at ~ 2.5 bar); compact, high heat release.

Supercritical technology

- **Critical point of water: 221.2 bar, 374.15 deg C.** Above it, no distinction between liquid and vapour -> no boiling, no drum.
- **Supercritical:** 240-250 bar, 540-565 deg C; **Ultra-supercritical (USC):** 270-300 bar, 600-620 deg C; **Advanced USC:** 700+ deg C (nickel alloys).
- Plant efficiency improves from ~ 35% (subcritical) to **40-42% (SC)** and **44-46% (USC)** -> less coal, less CO_2.
- Indian units: 660 MW and 800 MW supercritical sets (Sipat, Mundra, Barh, Sagardighi).
- * Challenges: high-temperature metallurgy, feed water purity, accurate control, thick walls (thermal stress).

4. Analysis of Power Plant Cycles

- Ideal cycle for steam plants = **Rankine cycle**: boiler -> turbine -> condenser -> pump.

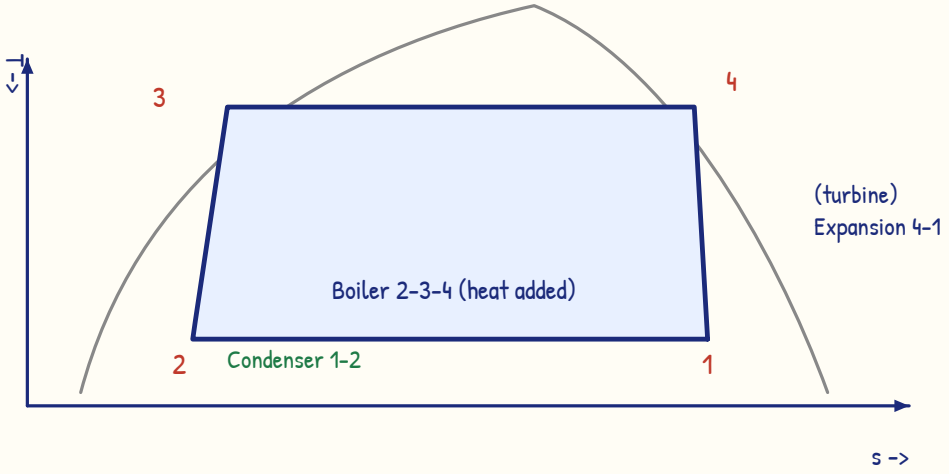


Fig: Rankine cycle on the T-s diagram (1: turbine exit, 2: pump exit, 3-4: boiler)

Processes

- 1-2: condensate pumping (adiabatic, reversible, small work). 2-3-4: heat addition in boiler (water heating, boiling, superheating at constant pressure).
- 4-1: **isentropic expansion** in turbine; 1 -> 2 : constant-pressure heat rejection in condenser (typically 0.04-0.1 bar abs).

Rankine cycle formulae
• Turbine work $W_T = h_4 - h_1$; Pump work $W_P = h_2 - h_1' = v \times (p_b - p_c)$

- Heat supplied $Q_1 = h_4 - h_2$; **Efficiency** = $(W_T - W_P) / Q_1$
- **Specific steam consumption** (kg/kWh) = 3600 / $(W_T - W_P)$; **Heat rate** = SSC x Q_1 (kJ/kWh)
- Back-work ratio = W_P / W_T (tiny, about 1%)

- Rankine efficiency increases with: **higher boiler pressure, higher superheat temperature, lower condenser pressure**.
- Drawbacks of raising pressure alone: wetter exhaust (low dryness fraction < 0.85) -> erosion of last-stage blades -> solved by **reheat**.

Reheat cycle

- Steam is taken out after HP turbine, **reheated** in the boiler to (say) 540 deg C, and expanded in IP / LP turbines. Raises dryness at exhaust and cycle efficiency by 4-5%.

Regenerative cycle

- Some steam is **bled** from the turbine to heat the feed water in **feed-water heaters** (open / closed type) -> the average temperature of heat addition rises -> efficiency rises; also helps deaeration.
- With ideal infinite stages, the efficiency approaches **Carnot**. Practical stations use **6-8 heaters**.

Reheat-regenerative cycle (modern practice)

- Combines both: superheat, **single reheat** (double in large units), 7-8 bleed points. Overall thermal efficiency 38-45%.

Binary vapour cycle and cogeneration

- **Binary cycle:** mercury (high temp.) + steam (low temp.) -- historic, mercury toxic. **Cogeneration (CHP):** back-pressure / extraction turbines supply process steam + power; fuel utilisation 70-85%.

5. Modern Trends in Cycle Improvement and Waste Heat Recovery

- Increase **steam parameters** (SC / USC), reheat, more regenerative heaters, low condenser pressure with good cooling.
- **Combined cycle (gas + steam)** efficiency 55-60%; **IGCC** (integrated gasification combined cycle) for coal.
- **Variable-pressure operation**, better turbine blade design, **digital controls** and condition monitoring, drives with **variable-frequency** drives for pumps and

fans.

Waste heat recovery (WHR)

- About 10–15% of energy is lost in flue gas and the rest in the condenser. Recovered by:
  - **Economiser** -- feed water heated with flue gas (every 20 deg C drop in gas temperature gives ~ 1% fuel saving).
  - **Air pre-heater** -- combustion air heated by flue gas (recuperative tubular type, **regenerative Ljungstrom** rotary type).
  - **Waste heat recovery boiler (HRSG)** in combined cycle and process industries.
  - Condenser heat for district heating / desalination; blow-down heat recovery; **steam jacketing** and insulation.
- Efficiency improvement of a boiler: **dry flue gas loss**, moisture loss and unburnt carbon loss are the main losses; boiler efficiency 85–90% (on GCV).

6. Fluidised Bed Boilers

- Coal (crushed, 6–10 mm) + limestone are supported on a **bed of ash / sand**; air from below at velocity 1–3 m/s makes the bed behave like a **boiling liquid** (fluidised) -> intense mixing.



Fig: Principle of a fluidised-bed boiler

Features

- Bed temperature **800–950 deg C** (below ash fusion point) -> **low NO_x** (thermal NO_x negligible).
- **SO_2 captured in the bed** by limestone ($CaCO_3$ -> CaO; $CaO + SO_2 + 1/2\ O_2$ -> $CaSO_4$) -- 90% removal with no FGD.
- Burns **low-grade, high-ash and multiple fuels** (coal washery rejects, biomass, petcoke, lignite). High heat-transfer, compact.

Type	Description
Bubbling FBC (BFBC)	Low velocity (1–3 m/s); distinct bed surface; in-bed tubes; for small units
Circulating FBC (CFBC)	High velocity (4–8 m/s); solids carried out, caught in cyclone and returned; suited to 125–300 MW

Type	Description
Pressurised FBC (PFBC)	Bed at 10–16 bar; hot gases drive a gas turbine -> combined cycle

- Drawbacks: higher auxiliary power (fans), erosion of tubes, limestone handling, N_2O emission.

7. Fuel and Handling Equipment – Types of Coals

- **Coal** forms from plants by coalification: Peat -> Lignite -> Sub-bituminous -> Bituminous -> Anthracite (carbon content and calorific value increase).

Type	Fixed carbon	Volatile matter	GCV (kcal/kg)	Remarks
Peat	40–55%	high	3500–4500	Early stage, 80% moisture
Lignite	60–70%	30–40%	4000–5500	High moisture, brown, burns easily
Bituminous	70–86%	15–40%	6500–8000	Best for power / coking, soft black
Anthracite	86–98%	3–8%	7800–8200	Hard, smokeless, hard to ignite

- Indian coals: mostly **non-coking, ash 30–45%**, moisture 4–10%, GCV about 3500–5500 kcal/kg.
- **Proximate analysis:** moisture, volatile matter, fixed carbon, ash. **Ultimate analysis:** C, H, O, N, S (% by weight). Higher calorific value (GCV) = LCV + latent heat of moisture.
- Dulong: **GCV = 33,800 C + 1,44,000 (H – O/8) + 9,270 S kJ/kg**.
- Fuel oil and natural gas are also used for start-up and flame support.

8. Coal Handling

- Large stations burn **10,000–20,000 tonnes of coal per day**; the handling system must be **reliable, continuous and dust-free**.

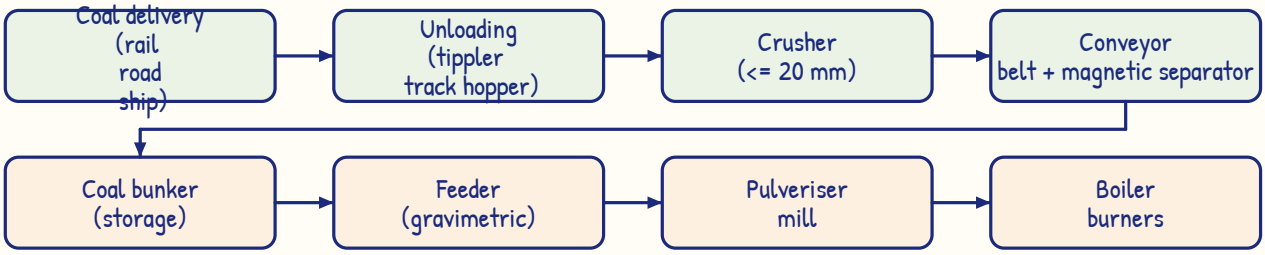


Fig: Coal handling sequence

Stages

- **Delivery:** railway wagons (BOXN), trucks, ships, MGR (merry-go-round) system.
- **Unloading:** wagon tippler, bottom-discharge hopper, car shaker, grab bucket crane.
- **Preparation:** crushers (jaw, hammer, ring-granulator), screens, **magnetic separators** (remove iron), sampling and weighing (**belt weigher**).
- **Transfer:** belt conveyor, screw conveyor, bucket elevator, skip hoist, chain / apron conveyor.
- **Storage:** bunkers (daily) and open stock-yard (live and dead storage).

Choice of handling equipment

Equipment	Merits	Limits
Belt conveyor	Large capacity (up to 5000 t/h), long distance, little power, quiet, cheap	Inclination up to ~ 18–20 deg only; dust
Screw conveyor	Simple, enclosed, low cost	Wear, short distance, small quantity
Bucket elevator	Vertical lift, compact	Limited capacity, wear
Skip hoist	Vertical / steep lift	Intermittent, higher cost
Apron / chain	Handles hot / lumpy coal, heavy duty	Costly maintenance

* Choice depends on: capacity required, distance and elevation, nature (size, moisture) of coal, space, capital + running cost and reliability.

Coal storage

- **Live storage** (for 1 week, near the boiler) and **dead storage** (30 – 60 days stock, compacted, kept as a reserve).
- Problems: **spontaneous combustion** (oxidation heating), loss of calorific value, wind loss, fire. Prevent by compacting in layers, side slopes, ventilation + temperature probes, keeping separate coal types, and **first in first out** use.

- Under-water storage avoids oxidation (cost + wetting issues).

9. Ash Handling Systems

- Coal with 40% ash gives huge ash: a 500 MW station may give **> 6000 t/day**. Ash = ***bottom ash*** (clinker, 20%) + ***fly ash*** (80%, in pulverised firing).
- Ash is ***abrasive, hot and dusty*** -- the handling system should be safe, efficient and economical.

System	Principle	Remarks
Mechanical	Belt, bucket or scraper conveyor	Small plants; high wear
Hydraulic	Ash sluiced with water to a sump / ash pond (slurry pump)	Common, large plants; large water demand, pollution
Pneumatic	Ash conveyed in pipes by vacuum (negative) or pressure (positive) air	Dry fly ash; clean; for ESP hoppers; long distance
Steam jet	Steam jet carries ash in a pipe	Small; high steam use

- ***Dense-phase*** pneumatic conveying and ***high-concentration slurry disposal (HCSD)*** are used in modern plants to save water.
- Utilisation: cement, bricks, concrete blocks, road embankments, land-fill, mine filling (Indian policy: ***100% utilisation***).

10. Combustion Process – Properties of Coal

- **Combustion** = rapid chemical reaction of fuel with oxygen, releasing heat. **Three T's: Time, Temperature, Turbulence.**
- Coal burns in stages: **drying -> devolatilisation (volatile matter burns) -> char (fixed carbon) burning -> ash.**

Basic combustion reactions

- $C + O_2 \rightarrow CO_2 + 33{,}800$ kJ/kg; $2C + O_2 \rightarrow 2CO$ (incomplete)
- $2H_2 + O_2 \rightarrow 2H_2O + 1{,}44{,}000$ kJ/kg; $S + O_2 \rightarrow SO_2 + 9{,}270$ kJ/kg
- **Theoretical air** for 1 kg coal = 11.6 C + 34.8 (H – O/8) + 4.35 S (kg of air). **Excess air** (20–30% for coal) is given for complete combustion.

Important properties of coal for burning

- **Calorific value**, volatile matter (easy ignition, long flame), fixed carbon, moisture (lowers CV, ignition difficult), **ash content and ash fusion temperature**, caking and swelling index, grindability (**Hardgrove index**), sulphur content, size.

11. Fuel Beds – Overfeed and Underfeed

- **Overfeed:** fresh coal is fed **on top** of burning bed, air from below; coal passes through zones -> ignition from the hot bed below. (travelling grate, spreader).
- **Underfeed:** fresh coal is fed **from below** the fire by a screw / ram; volatiles rise through the hot fuel bed above and burn completely -> **smokeless**. (retort stoker)

Stokers (mechanical firing)

- Used for boilers up to ~ 100–200 t/h of steam.

Stoker	Description	Features
Travelling grate	Endless chain grate moves from front to rear; coal on grate, air from below in zones; ash dropped at end	20–50 t/h; suits non-caking coal, low ash fusion; ignition arch needed
Spreader	Coal thrown by rotor / flipper onto grate; fine burns in suspension, coarse on grate	Quick response, handles many coals; fly-ash high, needs dust collector

Stoker	Description	Features
Retort (underfeed)	Coal pushed into retort by screw, rises, air through tuyeres; ash overflows on sides	Small boilers, bituminous caking coal; smokeless

- ***Vibrating grate*** and ***chain grate*** are variants. Stokers are limited to modest capacity -> for large units ***pulverised fuel firing***.

12. Pulverised Fuel Burning System

- Coal is ground to powder (***75% through 200 mesh = 75 micron***) and blown by ***primary air*** into the furnace where it burns in suspension like gas -> very high intensity.

Advantages

- Large boilers (> 100 t/h) possible, burns any coal (also low grade), quick response, low excess air (15–20%), high efficiency, less ash on grate.

Disadvantages

- High capital + power for mills, fly-ash and dust pollution (needs ESP), explosion risk, flame stability needs support oil, furnace slagging.

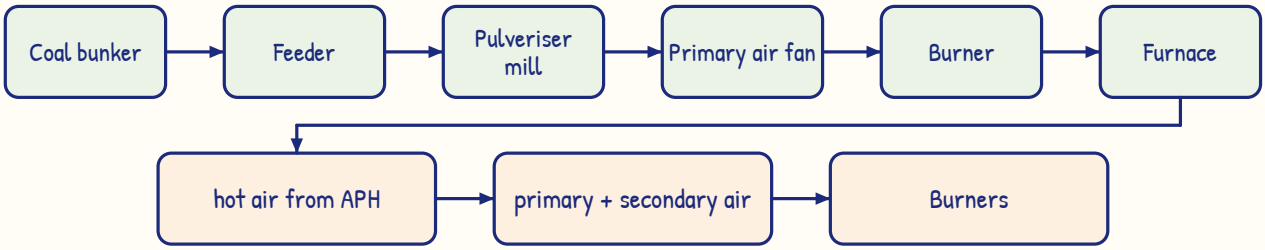


Fig: Pulverised-fuel firing (unit system)

Components

- ***Feeder*** (regulates coal flow), ***pulveriser / mill*** (ball and tube mill -- slow; ***bowl / ring-roll mill*** -- medium; ***hammer / impact mill*** -- high speed, for lignite), ***classifier***, ***primary air fan*** (carries + dries coal), ***burners*** (tangential, wall, or downshot), ***igniters***.
- ***Unit system*** (each mill feeds one boiler directly, common now) vs ***central (bin) system*** (powder stored in bunker, flexible).
- ***Types of firing:*** tangential (corner) firing -- fireball in the furnace; ***front / opposed wall firing*** with swirl burners.
- ***Slag*** removal: ***wet bottom*** (molten ash) and ***dry bottom*** furnace (ash as solid, common).

13. Cyclone Furnace

- A horizontal **water-cooled cylinder** attached to the boiler furnace; crushed coal (**4-6 mm**, not powdered) is fed into the cyclone with **primary air tangentially** (swirl).
- Secondary air enters tangentially at a high speed (~ 100 m/s) -> coal particles are flung to the wall and burn in a **molten slag layer**; heat release ~ 5 x 10^6 kcal / m^3h at **1650-1900 deg C**.
- **Advantages:** crushed coal, less fly ash (**~ 20-30%**; the slag is removed as liquid), small furnace volume, low excess air (10-15%), uses low-fusion-temp ash coals.
- **Disadvantages:** high fan power (high-pressure air), high NO_x, needs coals with low ash fusion temperature, high maintenance.

14. Combustion Needs and Draught System

- Combustion needs a **continuous supply of air** and removal of flue gas -> created by **draught** = pressure difference that causes flow of air / gas.

Types of draught

Type	Created by	Remarks
Natural	Chimney (density difference hot gas vs ambient)	Simple, no power; small; height 20-100 m; not controllable
Induced (ID fan)	Fan after boiler (pressure below atm. in furnace)	Handles hot gas; with chimney; leaks air inwards
Forced (FD fan)	Fan before boiler (furnace above atm.)	Cold air, small fan; hot gas may leak out
Balanced	Both FD + ID fan; furnace slightly negative (-5 to -10 mm WG)	**Used in large plants**; best control

Chimney draught

- Draught head: $h_w = H (\rho_a - \rho_g)$ in metres of gas column; in mm water: $h_w = 353\ H\ (1/T_a - 1/T_g)$ with temperatures in K.
- Maximum draught is obtained when $T_g = 2\ T_a$ (about); practical chimneys have flue gas at 150-200 deg C.

- Height of chimney is also fixed by the **pollution dispersion**: 220 m for > 500 MW units (see Unit I).

- **Fans:** centrifugal (backward curved / radial), **axial fans** for large flows (**with variable pitch**). Flow control: dampers, inlet guide vanes, speed (VFD).

15. Dust Collectors

- Remove **fly ash** from flue gas before the chimney.

Type	Working	Efficiency
Gravity settling chamber	Velocity reduced, dust settles	40–60% (coarse)
Cyclone separator	Centrifugal force in vortex	70–90%
Wet scrubber	Water spray captures particles	90–95%
Fabric (bag) filter	Dust retained on cloth bags	99%+ (fine)
Electrostatic precipitator (ESP)	Dust charged by corona (30–60 kV DC), pulled to collecting plates, rapped off to hoppers	**99–99.9%**

- ESP performance depends on the dust resistivity (10^{7}–10^{10} ohm-cm ideal), gas velocity (1 m/s), and plate area (**Deutsch** equation: eta = $1 - e^{-wA/Q}$).

16. Cooling Towers and Heat Rejection

- Condenser carries heat from exhaust steam (about 55–60% of fuel energy). The warm cooling water is cooled in a **cooling tower** and reused (closed cycle) if water is scarce.
- **Principle:** warm water falls through **fill**; part evaporates and removes latent heat -> cools the rest (can reach near the **wet-bulb** temperature).

Type	Features
Natural draught (hyperbolic)	Large shell 100–200 m, no fan, low running cost, high capital; large power stations
Mechanical draught (induced / forced fan)	Smaller, compact, fan power, controllable; cells in a row
Wet / Dry / Hybrid	Wet: evaporation; Dry: radiator (no loss of water, expensive); Hybrid: both, reduces plume

- Terms: **Range** = hot water temp. – cold water temp.; **Approach** = cold water temp. – wet bulb temp.; **Efficiency** = Range / (Range + Approach).
- Losses: evaporation (~ 1%), drift, blow-down. Make-up water ~ 2–3% of circulating water. Alternatives: cooling ponds, spray ponds, once-through (sea water).

17. Analysis of Pollution from Thermal Power Plants, Controls and CO_2 Recorder

- Pollution: **fly ash (SPM)**, SO_2, NO_x, CO, CO_2, mercury, hot water, noise and coal dust. A 500 MW station may emit ~ **10–15 Mt CO_2 per year** and burns over 7500 tonnes/day of coal.

Pollutant	Source	Control
Particulates	Fly ash	ESP, bag filter, cyclone
SO_2	Sulphur in coal	FGD, low-S coal, FBC with limestone
NO_x	High-temp. combustion	Low-NO_x burners, OFA, SCR / SNCR
CO_2	Combustion	High efficiency, CCS, co-firing biomass
Thermal	Condenser	Cooling tower

- Also: tall chimneys, coal washing, green belt, ash pond lining, ambient air monitoring (see Unit I).

CO_2 recorder

- Used for **flue gas analysis** to check combustion efficiency: CO_2 % is high when excess air is low; the flue-gas CO_2 percentage and the O_2 % tell the **air–fuel ratio**.
- **Orsat apparatus** (chemical absorption): measure a flue-gas sample (100 cc) over water, pass through absorbents: **KOH** absorbs CO_2, **alkaline pyrogallol** absorbs O_2, **cuprous chloride** absorbs CO. The decrease in volume = % of each gas.
- **CO_2 recorder (automatic):** a gas sample is continuously drawn; CO_2 is absorbed in KOH solution -> volume decrease measured and recorded on a chart; based on this the operator adjusts air supply. Modern units use **IR / zirconia O_2 analysers**.
- Typical values: CO_2 = 12–15% in coal flue gas; **Excess air % = $21/(21 - O_2) - 1$** (in %).

Quick revision

- Rankine efficiency up by higher pressure, superheat, reheat, regeneration, lower back-pressure.
- FBC: 800–950 deg C, low NO_x, in-bed SO_2 capture. PF firing: 75 micron, > 100 t/h. Cyclone: crushed coal, slag.
- Balanced draught = FD + ID. ESP 99%+. Cooling tower approach = cold water – WBT.